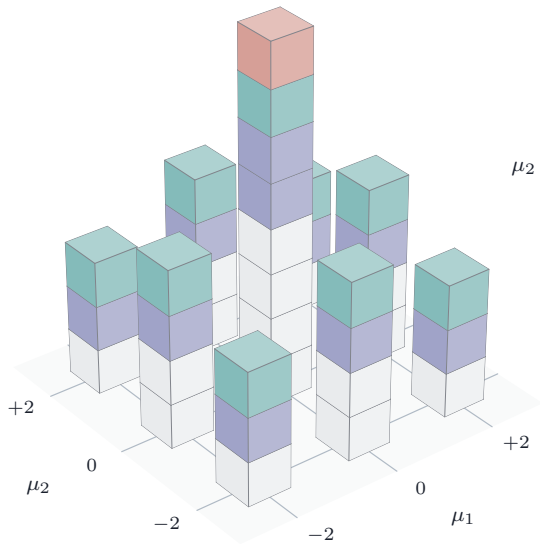




		μ_1		
		-2	0	+2
μ_2	+2	$d = 3$ $\frac{1}{2} P_1 P_2 Q_1$	$d = 4$ $\frac{1}{4} P_1 P_2 Q_1 Q_3$	$d = 3$ $\frac{1}{2} P_1 P_2 Q_1$
	0	$d = 4$ $\frac{1}{4} P_1 P_2 Q_1 Q_2$	$d = 8, \ker = \mathbb{C}\kappa$ $\frac{9}{256} R P_1 P_2 Q_1 Q_2 Q_3 Q_4$	$d = 4$ $\frac{1}{4} P_1 P_2 Q_1 Q_2$
	-2	$d = 3$ $\frac{1}{2} P_1 P_2 Q_1$	$d = 4$ $\frac{1}{4} P_1 P_2 Q_1 Q_3$	$d = 3$ $\frac{1}{2} P_1 P_2 Q_1$

tangent space at Y of the Hodge locus of ω_a

- ω_a exceptional: $\mathbb{C}\kappa$, dim 1
- $a = (3, 5, 5, 5)$: dim 10
- $a = (1, 1, 1, 1)$: dim 20
- $T_Y \mathfrak{S} = \mathfrak{sp}^{-1,1}$: dim 36

$$\begin{aligned}
 2P_1 &= \sum_{i < j} (a_i - a_j)^2, & 2P_2 &= \sum_{i < j} (a_i + a_j)^2, & R &= a_2^2 + a_3^2, & Q_4 &= (3a_1 - a_0 - a_2 - a_3)^2 + 16(a_2 - a_3)^2 \\
 33Q_1 &= 3(11a_1 - a_0 - 5a_2 - 5a_3)^2 + 8(6a_2 - a_0 - 5a_3)^2 + 88(a_3 - a_0)^2 \\
 119Q_2 &= 21(17a_1 + a_0 + 5a_2 + a_3)^2 + 16(7a_2 - 2a_0 + 15a_3)^2 + 272(3a_3 + a_0)^2 \\
 51Q_3 &= 9(17a_1 + a_0 + a_2 + 5a_3)^2 + 8(18a_2 + a_0 + 5a_3)^2 + 136(a_3 - a_0)^2
 \end{aligned}$$